Feasibility of Digital Health Services for Educating the Community People Regarding Lifestyle Modification Combating Noncommunicable Diseases

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Abstract. Mobile health (mHealth), or the use of mobile technology to improve health, is a rapidly expanding field. There have been a number of intervention studies based on mobile apps and most of these focused on specific medical issues. A majority of the available health apps are for health and wellness promotion and disease prevention for the general public. But the pace of traditional academic research is slow and less nimble relative to commercial app development resulting in huge lags in dissemination into commercial settings for public health. This paper assessed the content, usability and efficacy of the commercially launched digital healthcare platform developed by CMED Health for the purpose of preventing major noncommunicable diseases in the rural communities of Bangladesh. A combining approach of quantitative, qualitative and participant observation were used in this study following the principles and process of Action Research methodology. The study evaluated the CMED health app by content analysis, usability testing, health observational study, and efficacy testing. The view of the service receivers, service providers and health managers of the concerned organization was evaluated to test the feasibility. The quantitative data was collected using a semi-structured questionnaire from 393 service receivers and qualitative data was collected through Key Informant Interview (KII) of key-informants of service providing team, senior staff members and village opinion leaders and Focus Group Discussions (FGDs) with the direct service providers and service receivers. The study

Q1+Q2

Q3 People

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analyzed quantitative health data generated from the service delivery during 6-month piloting with the members of more than 7,000 households in 21 villages. Q3 People
Q4 The main reasons for accepting the CMED health app was showing the results instantly on the screen, instant advices and getting health services in a smart system within a short period of time. This also increased their trust on the health care providers and they were benefitted by the health records stored in the cloud of the user for their next visit to physicians and getting medicines. The people
Q4 also mentioned about the suitability of the digital healthcare services for preventing diabetes and hypertension in the community.

Keywords: Mobile health app · Usability · Noncommunicable diseases · Prevention · Health records · Smart device

1 Introduction

Mobile health (mHealth), or the use of mobile technology to improve health, is a rapidly expanding field [1]. As of 2015, more than 165,000 mHealth apps were available on the Apple iTunes and Android app in play stores, and 34% of mobile phone owners had at least one health app on their mobile device [2–4]. Commercial mobile apps for health behaviour change are flourishing in the marketplace. Although health apps have drawn great public interest and use, little evidence exists about the usability and efficacy of the majority of commercially available apps [5, 6]. Producing an app for public use requires content, programming, design expertise, the ability to continually host and update the app, and the resources to provide both customer service and technical support [7–9].

CMED Health is an Internet of Things (IoT) enabled Artificial Intelligence (AI) driven cloud based preventive and primary health care service provider [10]. CMED uses portable smart medical sensors and devices connected to a smart phone application for the measurement of vital signs or health parameters and store data to its secured cloud server. Based on acquired data, CMED clinical decision support systems (CDSS) perform health risks assessment and provide instant feedback to the user about their health status, and moreover, it connects to a doctor if he/she has risks. CMED digital healthcare platform also generates sharable health reports from stored health data which helps doctors to minimize diagnostic time and to provide better treatment. The architecture of CMED digital healthcare platform for providing preventive and primary healthcare services is shown in Fig. 1.

Unlike the existing systems used usually, CMED system keep records of the user's data automatically. It also give suggestions to the users to consult with the physician when their health condition is at risk. At this current stage, CMED system can monitor the vital parameters of human health like blood pressure, blood glucose, pulse, blood Q3 Feat oxygen saturation (SpO2), body temperature, weight, height and body mass index (BMI) [10]. The pictorial representation of CMED Smart devices, health kit, and App showing color coded measurement records as well as example of measuring blood glucose using smart devices, App and CDSS feedback is shown in Fig. 2.

Over the last two decades, Bangladesh, a lower middle-income country, has experienced a rapid demographic and epidemiological transition [11]. The population is

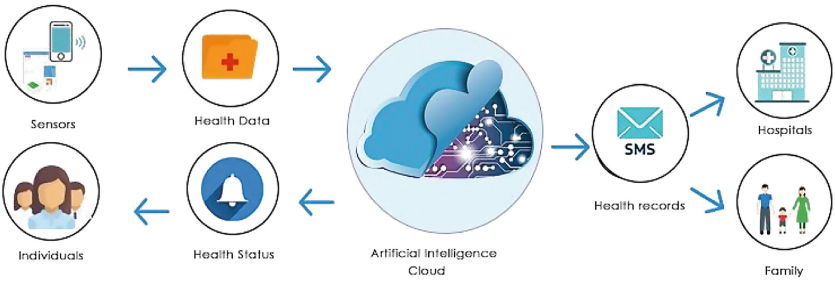
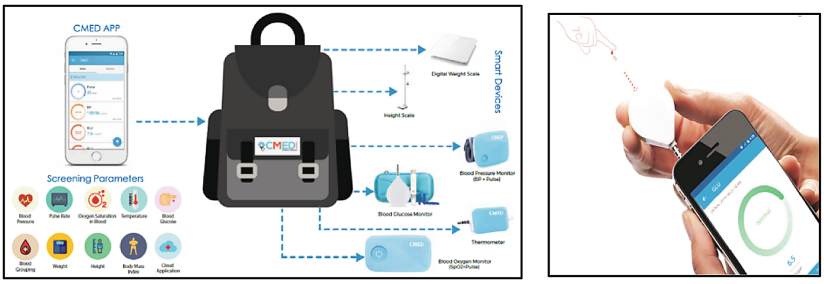


Fig. 1. Architecture of CMED IoT enabled Artificial Intelligence driven Cloud based Preventive and Primary Healthcare Platform, which includes CMED smart devices, mobile App, CDSS, and Cloud platform.



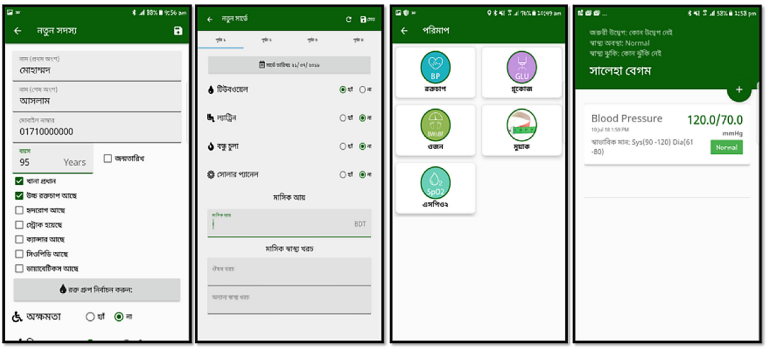
(a) (b)

Fig. 2. (a) Smart devices for screening health parameters (blood pressure, blood glucose, pulse, blood oxygen saturation (SpO2), body temperature, weight, height and body mass index) equipping in CMED health kit, and CMED App showing color coded measurement records; and (b) example of measuring blood glucose using smart devices, App and CDSS feedback based on the blood glucose value in the CMED Digital preventive and primary healthcare platform.

projected to increase substantially from 147 million in 2007 to 218 million in 2050 [12]. In 1986, the total deaths due to chronic diseases was 8% which increased to 68% by 2006 in rural Bangladesh [13]. At least in part, this can be explained by changes in nutritional status. People in Bangladesh are not conscious about their health. Bangladesh is actually facing a major crisis in this sector. Being one of the highly populated country, a number of people are suffering due to obesity, cardiovascular diseases, diabetes and many other non-communicable diseases. In a nationally representative study among adults (>35 years) 18.9% were overweight and 4.6% were obese [14] whereas about 17.9% school children were obese [15], around 21% people are suffering from hypertension [16], and more than 10% are diabetic [17]. On the other hand, due to the unhealthy dietary habit, physical inactivity and other behavioural factors health risks are increasing simultaneously. According to World Health Organization, 67% of the deaths occurs in low-and middle-income countries due to non-communicable diseases with 31% of these deaths is due to cardiovascular diseases and 3% is due to diabetes [18]. Best way to deal with non-communicable disease is

preventive healthcare. Research has claimed that preventive healthcare can reduce the risk of NCDs – upto 90% strokes, 80% heart attack and 73% diabetes can be prevented by changing lifestyle with healthy diet and physical exercise [19–21]. Thus preventive healthcare with regular health monitoring and lifestyle modification can minimize the healthcare risk, cost and time of an individual and of the nation as a whole [22].

Q2 CMED Health Platform is designed to reduce the risk of life threatening non-communicable diseases like stroke, heart attack, diabetes, obesity, etc. With the regular health monitoring systems, users would be able to reduce diagnostic time, follow up cost, cost of hospitalization and hospital days significantly. Community Health Workers (CHW) can also play a vital role in the rural areas by using CMED systems. This paper Q1+Q2 assessed the content, usability and efficacy of the commercially launched digital healthcare platform developed by CMED Health for the purpose of preventing major noncommunicable diseases (cardiovascular diseases, hypertension and diabetes) and improving primary health care in the rural communities of Bangladesh. Figure 3 showing CMED App user registration process with demographics and household information collection, health parameters screening using App and smart devices and measurement records of the user (a); and CMED Digital Healthcare Platform Dashboard



(a)

(b)

Fig. 3. (a) CMED App showing user registration process with demographics and household information collection, health parameters screening using App and smart devices as well as showing color coded measurement records of the user. (b) CMED Digital Healthcare Platform Dashboard showing aggregated information and statistics on household visits, health conditions, household and demographic status, health workers work performance and so on.

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2 Methods

A triangulation of research methods combining quantitative, qualitative and participant observation were used in this study following the principles and process of Action Research Methodology. The study was conducted through a pilot project on digital health services implemented with more than 32,000 people in about 7,000 households of 21 villages in Dhamrai upazila (sub-district) of greater Dhaka district in Bangladesh. The study was conducted during the period of July to December of 2019 with joint initiative of CMED Health; AIMS Lab, Department of CSE, United International University; Department of NCD, Bangladesh University of Health Sciences; Palli Karma Sahayak Foundation (PKSF) and local Non-Government Organization (NGO), SOJAG. Demonstration of digital healthcare services in rural areas of Bangladesh using CMED preventive and primary healthcare platform is shown in Fig. 4. The study evaluated the digital healthcare services model implemented through mobile health apps developed by CMED Health. It includes content analysis, usability testing, health observational study, and efficacy testing as discussed below. Q3 Feat



Fig. 4. Demonstration of digital healthcare services delivery using CMED preventive and primary healthcare platform in Dhamrai upazila, a rural areas of Bangladesh.

(a) The contents were evaluated by comparing the cut off values and interpretation with the National Protocol on Diabetes and Hypertension. The contents of the CDSS were evaluated by the stakeholders such as medical professionals, health service providers, health service receivers, service supervisors and management staff members of the NGO. As a guiding standard, the content evaluation study followed the national protocol in the rural setting for pilot implementation.
(b) The health information and advices incorporated in the digital healthcare services model were validated by expert consultation through arranging a workshop at the policy level.
(c) During piloting, the usability testing of the services model has been conducted. Through a testing-and-fixing process the interfaces of the mobile health service app were established. The field health service providers and the management staff members were used for this usability testing in the working environment. It has

been assessed how well the app functions and serves its intended purpose for a target population which was explored through a piloting or field testing.

(d) The efficacy of the digital health app was evaluated through comparing the measurement values with the manual and traditional methods. Through 6-month piloting the efficacy of the app and underlying systems has been evaluated in the socio-cultural setting of rural Bangladesh.

Q3 People The rural health service providers provided health services to the every household members of 21 villages using the CMED digital healthcare platform for 6 months. The study team observed the participants, the process of providing health services, and the use of smart devices and mobile app on a continuous manner, and went through a process of evaluation. The study team collected quantitative and qualitative data from (1) Rural Health Service Providers, (2) Senior Management Staff Members, Q3 People Q3 People (3) Selected Village Opinion Leaders, and (4) Service Receivers. And the feasibility of the digital health services was evaluated by taking the view from the service receivers, providers and health managers of the concerned organization.

The study analyzed quantitative health data generated from the service delivery during 6-month piloting with the members of more than 7,000 households in 21 villages. The quantitative data was collected using separate semi-structured questionnaire from 393 service receivers of the piloting area. Qualitative data has also been collected through Key Informant Interview (KII) of key-informants of service providing team, senior staff members and village opinion leaders and conducting four Focus Group Discussions (FGDs) with the key participant groups of the pilot project. Two FGDs were conducted with the service providers who used both the CMED smart devices and app, and another one was conducted with the group who used manual devices and app for providing digital services. Two more FGDs were conducted among the service receivers in two different villages. Prefixed separate FGD guidelines were used for conducting FGDs among the service providers and receivers. The discussion was recorded in a tape recorder and also noted in pen and paper. The transcription of the discussion findings has been done and presented as speech mark in the result. Figure 5 showing Focused Grouped Discussion on digital health service with the rural people (service receivers) and the health workers (service providers).

Q3 People



Fig. 5. Focused Grouped Discussion on digital health service with the rural people (service receivers) and the health workers (service providers).

3 Results

During the CMED digital health services pilot project, 17,907 adult population out of 32,122 have received the primary health screening services through the use of CMED digital platform and smart medical devices. The services include regular blood pressure monitoring, glucose monitoring, BMI, SPO2 and MUAC for child nutrition measurement of aged under 5 years. Q3 People

Table 1 shows the number of services provided in different measurements during pilot study. Of the 17907 people, blood pressure, BMI and SPO2 have been measured for the highest number of times among all other health screening.

Table 1. Number of measurements taken in each category during pilot study.

Measurement type	BP	Glucose	BMI	SPO2	MUAC
Number of measurements	36326	1348	34721	38400	1229

3.1 Quantitative Findings

Gender of the Respondents from Service Receivers. The responses were taken at the day time when most of the men remain busy with their works outside. As a result, most of the respondents were female, which consists of 92.10% female and 7.90% male. Q3 People

Types of Services Received. Figure 6 shows the primary health screening services taken using CMED digital healthcare system as percentage of survey population. On average, more than 90% of the adult population have taken the primary health screening services. On the other hand, 21% children of the total population have received MUAC test to measure malnutrition.

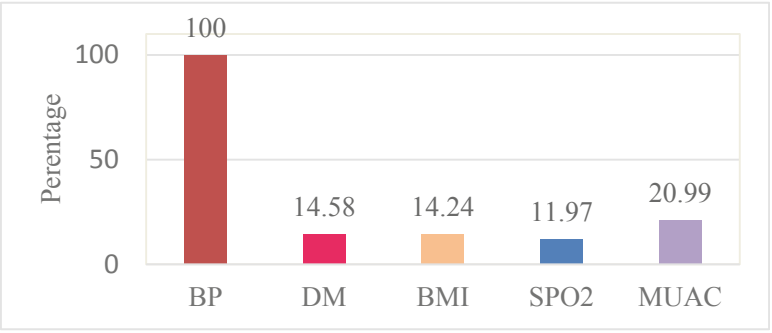


Fig. 6. Type of services and its percentage received by the community people using Digital Healthcare Platform.

Acceptance and Preference. About 99.5% of the respondents exhibited satisfaction when asked about the acceptance of CMED digital health services. The respondents Q4

also expect continuation of the CMED digital health services in future. Almost all respondents could mention about the services they received from the local NGO (SOJAG) health workers including blood pressure, blood glucose, SPO2, MUAC, height, weight and BMI measurements. Most of the respondents told that the new digital method (CMED health app with smart devices) is better than the old manual method. The study found that majority of the service receivers accepted the service delivery model using smart devices and app. However, a very few service receivers recommended for the old manual machines as they are used to it and thought those are more reliable. When we explored the reasons of liking the new digital system, it was observed that more than quarter (27.3%) of the respondents mentioned new method is digital and smart looking, 14.7% of the respondents mentioned the results can be seen instantly on the screen, and can be compared with previous record (14.4%), about 13% respondents liked it as they can get the advices instantly through the app as shown in Fig. 7. Rest of the participants/population had different views like they are getting the accurate health results in a short time at their doorsteps. All the respondents told that Q4 the digital method using smart device and app is suitable for providing health services at the community level. A minor proportion of the respondents (1.6%) mentioned about the reasons of their dissatisfaction as they do not get the services and instant medicine completely free which is out of scope of the study and completely depends on the providers service model.

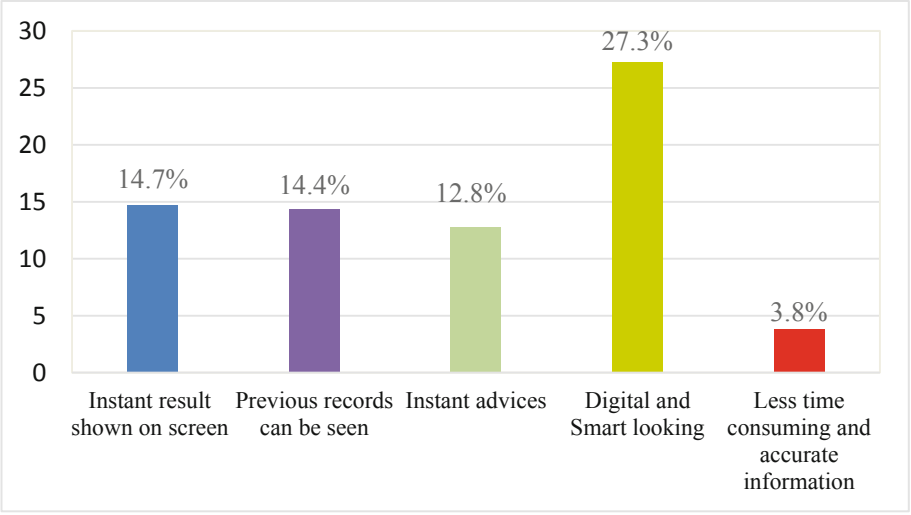


Fig. 7. Reasons for acceptance of the CMED digital healthcare platform and its percentage based on survey questionnaires.

We found that the majority of the sample liked CMED system in their regular health monitoring services for many reasons. Previously, they could not see the results with their own eyes. All they had to do is to rely on what the health workers say to them. As a result there was a certain level of understanding gap. On top of that, they could not

remember the records provided by the health workers or lost the written reports. Now they can see the reports on the screen with their eyes instantly which they can believe. They now do not have to worry about maintaining their previous records as well. In addition, they find the system is very efficient and effective with the advices provided by the CMED CDSS. Based on the advices provided by the app, they are acting upon it by doing lifestyle modification with various exercises such as started walking or walking more than before. Q5

Usability of the App. All the respondents responded positively when asked about the usablity of CMED health services in the community level. They were satisfied with the interfaces and output formats of the app. Also regarding detection of the heath conditions, the smart app and device was more effective as it identified more diabetic and hypertensive patients in the community. For example, only 2.7% of the population knew that they have diabetes whereas 15.8% were found with diabetes after blood glucose measurement. Q4 After this finding, people have been found to be more concerned about the NCDs and they are acting upon to prevent it. For example, some of them started to exert physical efforts to stay healthy and fit. This has been possible due to the visibility of results on the phone's screen in native language, which they can understand easily. On the other hand, only 6.5% of the population knew that they have hypertension while after screening, 11.5% people were identified as hypertensive. Q4 Accordingly, more adult population who came to know their health condition through primary health screening services using CMED digital system can now reduce the risk of stroke, cardiovascular diseases and other related NCDs through life style modifications.

3.2 Qualitative Findings

FGD Findings from the Health Visitors. The health workers of the local NGO are working in this field for around six years and provide primary health services to the community people. These primary health services includes blood pressure monitoring, blood glucose monitoring, height measurement, weight measurement, SPO2 monitoring and MUAC measurement for the children who are aged under 5 years. Besides, they arrange two days of satellite clinic service in every week and help the rural people consult with the doctors in real time. Also they set various health camps for the villagers such as eye health camp, heart camp, ENT camp etc. Each health visitor covers at least 25 households every day. The FGD findings of the different health workers are presented as below-

> *"Previously we didn't measure the height and weight of all people. We didn't monitor BMI of the people as manual BMI calculation is very complicated and time consuming, which was very hard to do at the field level. However, now we can monitor BMI through the CMED system very easily."*

> *"Now I can give proper advice to the villagers using CMED app and smart devices for which they respect us more than before."*

"Now the villagers can see their blood pressure status directly on the screen through CMED health app, which helped them to trust the health workers more than before. Also they want to know the previous record of blood pressure, weight etc. and compare with the current measurements. With all these facilities, the villagers are satisfied and excited."

When they were asked about the usability of the app in measuring blood pressure and glucose at the field level they told that smart device and app is very suitable to use in the field. It will be helpful for the villagers. Now the villagers are more aware of their health condition. They are now satisfied as the health workers provide services using digital technique. More comments from health workers on digital services are presented below.

"The messages are very useful to the villagers and they try to obey their advices."

"The machine can say whether one is diabetic or not, hypertensive or not, overweight or not. As a result, the villagers are now more eager to test their primary health vitals. Also the result accuracy of the devices is more than 90 percent."

The health workers also mentioned that relatives of the piloting villages regretfully say that,

"You have so many advantages in your village. You get the health services in modern technique, but we don't have that service in our village. We didn't see it before. We also want this service in our village".

One of the villagers told to a health visitor,

"After measuring my weight, when I saw it was more than normal, I started walking. Now I walk regularly and feel better than before".

FGD Findings from the Service Receivers. The people in the village, who received digital healthcare service and participated in the FGD mentioned that,

"Previously when the health workers measured the blood pressure or other parameters, only we could not see it, but now in digital system we can see it directly on the screen and we get the suggestions at the same time. As we couldn't see the results before, we hardly trusted them thinking that could be wrong."

"We can take our previous records from the app and use it when we go to the doctors for consultation."

About the feasibility of the CMED app the villagers mentioned that,

"We think this health services provided by the smart device and app would be feasible to implement at the field level for detection of hypertension and diabetes among the community."

4 Discussion

In our country, 67% deaths occur due to noncommunicable disease like, heart attack, stroke, diabetes, cancer and chronic obstructive pulmonary disease and our health system is not ready to handle this soaring burden [23]. Cardiovascular (CVD) mortality

raised from 17% in 2014 to 30% in 2018 and 85% of these deaths are due to heart attack and stroke [24]. Still the people do not know their health status and there are about 36 lacs undiagnosed cases of diabetes in Bangladesh [22]. So people come to the physicians for solution with complications leading to increased morbidity, mortality and health expenditure. According to Bangladesh National Health Accounts, our out-of-pocket health expenditure is one of the highest (67% of the country's total health costs) in South-East Asian Region. Seventy percent of the health expenditure is spent on drugs, health care services, and curative health care while just 10% accounts to preventive care [25]. Still more than 70% people in our country live in rural area where health service delivery is not within the reach of the general people [26]. In these circumstances, we need a cost-effective health intervention which is acceptable and available at the community level. To make the health services accessible to the people CMED Health launched a digital healthcare services which can identify the health conditions at an early stage with low cost. Thus CMED Health is bringing upon change Q5 and creating impact in the health sector by saving lives. However, the actual behaviour change in the community couldn't be measured due to short duration of the study.

Regarding feasibility we have identified the following advantages of CMED digital healthcare platform-

1. Through this smart app health managers can constantly supervise the health workers and easily identify the gap in the system.
2. Due to the collection and storage of information in the digital system the usability of the app has been increased and now it is possible to do the research in various directions like socio-economic, gender based, environmental, disease risk factors identification and association with different factors.
3. Health administrators and policy makers can visualize the overall information of a population in a defined geographical area sitting in one place through the dashboard of the platform, like the total birth, death, marriage, pregnancy, disease prevalence, sufferings, medical expenses etc.
4. Collecting and recording the information at the same time has reduced the additional hassle for further data collection.
5. As the digital system works in offline, so the health workers could work uninterruptedly and finish their work in time even though the areas doesn't have good internet coverage.
6. Using this app, the workload of the health workers has been reduced a lot and their interest in work has been increased. As a result, the efficiency of the organization is also increasing.
7. Q5 The knowledge of the community people regarding their health risks and its prevention has been increased as instant advices are provided through the CDSS of the app during health monitoring.

CMED health created a cloud based digital healthcare platform through which specific health interventions can be designed against a particular health condition among a group of population (gender, age based). Also the app can be used to design a health insurance policy for the people based on their socio-economic characteristics.

5 Conclusion

As per opinion of the service providers and service receivers the CMED digital preventive and primary healthcare system is suitable and feasible to use at the rural and urban community level throughout the country. It was noticeable that awareness has been developed among the villagers after getting the health services through smart device and app with CDSS. So this digital method can be used as the primary prevention intervention model to prevent the exposure to risk factors of NCDs and reduce the burden of chronic diseases in a low resource setting like Bangladesh where there is a dearth of healthcare services in remote areas.

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